

Topic: Recursion, Dictionaries

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① Check My Understanding

Questions completed: 4/ 5

Answers revised after AI hints: 5 / 5

② My Misconception

Before: I thought...

Recursion was just a function calling itself, and dictionaries were a way to store some data

Now: I understand...

Recursion requires a base case to stop the function when it eventually finish.

Dictionaries are useful because they organize information using key-value relationships and it is very efficient to look up by key.

③ Challenge the AI

One AI-generated question I challenged:

How can you tell whether a problem is naturally suited for recursion rather than iteration?

Why?

☐ Ambiguous

☐ Oversimplified

☒ Technically questionable

☐ Too easy

☐ Other: _____

Brief explanation:

There is not always a clear boundary between recursion and iteration. Many problems can be solved effectively using either approach, so the answer may depend on factors such as readability, efficiency, and problem structure. Because of this, the question may not have a single correct answer.

④ One-Minute Reflection

One thing I am still unsure about:

I am still unsure about how dictionaries and recursion can be used together to solve more complex problems, especially when the dictionary is used to store or update information during each recursive call.